

# Notes: Bäcker-Peral, Hazell, and Mian (AER, 2026)

## Dynamics of the Long-Term Housing Yield: Evidence from Natural Experiments

### Contents

|          |                                                                              |          |
|----------|------------------------------------------------------------------------------|----------|
| <b>1</b> | <b>Big picture</b>                                                           | <b>3</b> |
| <b>2</b> | <b>Data and measurement (key objects)</b>                                    | <b>3</b> |
| 2.1      | Administrative lease extension panel . . . . .                               | 3        |
| 2.2      | Transaction prices, private listing data, and rents . . . . .                | 4        |
| 2.3      | Sample restrictions and timing notation . . . . .                            | 4        |
| 2.4      | Unobservables and economic objects . . . . .                                 | 4        |
| <b>3</b> | <b>Models and empirical specifications (all equations + interpretation)</b>  | <b>5</b> |
| 3.1      | The long-term housing yield . . . . .                                        | 5        |
| 3.2      | Pricing a finite-duration leasehold with an extension option . . . . .       | 5        |
| 3.3      | Difference-in-differences estimand . . . . .                                 | 5        |
| 3.4      | From DiD to the pricing equation . . . . .                                   | 6        |
| 3.5      | The DiD-NLLS estimator . . . . .                                             | 6        |
| 3.6      | Why the estimator differences out the short end . . . . .                    | 6        |
| 3.7      | Control-group construction using repeat sales . . . . .                      | 7        |
| 3.8      | Validation regression using rent growth . . . . .                            | 7        |
| 3.9      | Cross-sectional regressions for climate risk and supply elasticity . . . . . | 7        |
| <b>4</b> | <b>Identification and instruments (what makes the estimates credible)</b>    | <b>8</b> |
| 4.1      | What fails in naive approaches . . . . .                                     | 8        |
| 4.2      | What identifies the model in the new approach . . . . .                      | 8        |
| 4.3      | Why the parallel-trends assumption is plausible . . . . .                    | 8        |
| 4.4      | Why the repeat-sales control strengthens the design . . . . .                | 9        |
| 4.5      | What the instrument does in the supply-elasticity exercise . . . . .         | 9        |
| 4.6      | Relation to the literature . . . . .                                         | 9        |

|          |                                                                                     |           |
|----------|-------------------------------------------------------------------------------------|-----------|
| <b>5</b> | <b>Tables and figures</b>                                                           | <b>9</b>  |
| 5.1      | Figure 1: Time series of the expected long-term housing yield . . . . .             | 9         |
| 5.2      | Table 1 and Figure 2: Extension distribution and the timing notation . . . . .      | 10        |
| 5.3      | Figures 3 and 4: Why the estimator isolates the long end . . . . .                  | 11        |
| 5.4      | Table 2, Table 3, and Figure 5: Validation of parallel trends . . . . .             | 11        |
| 5.5      | Figure 6 and Figure 7: Event-study and NLLS intuition . . . . .                     | 12        |
| 5.6      | Table 4 and Figure 8: Average $y$ -star and the flat long-run yield curve . . . . . | 13        |
| 5.7      | Figure 9: Monthly real-time estimates of $y$ -star . . . . .                        | 13        |
| 5.8      | Figure 10: Stability relative to the cross-sectional approach . . . . .             | 13        |
| 5.9      | Figure 11: Comparing other asset yields to $y$ -star . . . . .                      | 14        |
| 5.10     | Table 5, Table 6, and Figure 12: Climate risk and supply elasticity . . . . .       | 15        |
| <b>6</b> | <b>Short summary</b>                                                                | <b>15</b> |
| <b>7</b> | <b>Conclusion</b>                                                                   | <b>15</b> |
| 7.1      | Core conclusion (what the paper establishes) . . . . .                              | 15        |
| 7.2      | Economic intuition . . . . .                                                        | 16        |
| 7.3      | Takeaway . . . . .                                                                  | 16        |

# 1 Big picture

- **Question.** How can we estimate the market’s expected **long-term housing yield**, denoted  $y_t^*$ , in a way that is insulated from short- and medium-run housing and macro shocks?
- **Setting.** The paper studies the United Kingdom leasehold housing market, where many apartments are sold with long leases and some leases are later extended by 90 years or more.
- **Core challenge.** Long-term yields are hard to observe directly because market prices reflect a mixture of short-run and long-run forces, while the underlying service flow of housing is itself difficult to measure.
- **Main idea.** Lease extensions create a natural experiment in asset maturity. Comparing the same property’s value before and after extension, relative to similar nonextending leaseholds, isolates the value of very long-horizon cash flows.
- **Main empirical strategy.** The paper estimates the gain in market value from lease extension with a **difference-in-differences** design, and then maps that gain into  $y_t^*$  using a discounted-cash-flow pricing equation estimated by **nonlinear least squares**.
- **Main results.** The expected long-term housing yield is around **5.2%** in the early 2000s, begins to fall around the Great Recession, and reaches about **2.7% in 2024**. The decline is larger in supply-inelastic places, while the level of  $y_t^*$  is higher in areas more exposed to long-run climate risk.
- **Interpretation.** The broad decline in  $y_t^*$  looks similar to long-run trends in other asset yields, which suggests a common driver such as a fall in the long-run safe rate  $r_t^*$  rather than something uniquely specific to UK housing.
- **Why this paper matters.** It provides a public, real-time, market-based measure of long-run expected yields using microdata and natural experiments, without imposing a large structural macro model.

## 2 Data and measurement (key objects)

### 2.1 Administrative lease extension panel

- **Institutional setting.** In the United Kingdom, a freehold is perpetual ownership of property, while a leasehold is a long-duration claim that can be bought, sold, and later extended.
- **Core data construction.** The paper combines closed lease titles and current open lease records from His Majesty’s Land Registry (HMLR) to identify historical lease extensions. This is the key innovation because public records ordinarily show only currently open leases.
- **Scale.** The authors identify roughly **319,000 leasehold extensions since 2000**. For the main treatment sample, they observe **143,614** extensions with transaction prices both before and after extension.
- **Usable DiD sample.** Using their repeat-sales control construction, they can form controls for **130,971** of the 143,614 treatment observations.

- **Extension lengths.** The most common extension is **90 years**, but a large share of extensions are for **700+ years**. In the treatment sample, there are 46,265 ninety-year extensions, 59,403 700+-year extensions, and 37,946 other extensions.
- **Timing.** Lease registration became mandatory in 2003, so extensions after that date are comprehensively recorded.

## 2.2 Transaction prices, private listing data, and rents

- **Transaction prices.** HMLR publishes all property transactions in England and Wales from 1995 onward, including the exact date, price, and address.
- **Hedonics and rents.** The authors augment the public data with Rightmove and Zoopla listing data, which contain bedrooms, bathrooms, floor area, property condition, and rents.
- **Role of private data.** These private data are not needed for the main estimate of  $y_t^*$ . They are used to validate identification by checking hedonic balance, renovation patterns, and rent growth.
- **Coverage.** About **80%** of transactions are matched to listing data, and rental data are available for roughly **40%** of properties.

## 2.3 Sample restrictions and timing notation

- **Timing notation.** A property is purchased at time  $t - h$ , sold at time  $t$ , and extended at time  $t - u$  where  $0 < u < h$ .
- **Lease duration notation.** If the lease would have had remaining duration  $T$  at sale absent extension, then it has duration  $T + h$  at purchase and duration  $T + k$  at sale after a  $k$ -year extension. The paper’s benchmark case is  $k = 90$ .
- **Flippers removed.** Properties with holding period  $h \leq 2$  years are excluded because quick purchase-renovate-resell cases may violate the identifying assumptions.
- **Other restrictions.** The sample excludes very short extensions, very short remaining lease durations, and extreme price-growth outliers.

## 2.4 Unobservables and economic objects

- **Service flow.**  $R_t$  is the current service flow or rental value of housing.
- **Yield components.** At horizon  $u$ , the housing yield is the sum of the risk-free rate and housing risk premium minus rental growth.
- **Long-run yield.** The main object  $y_t^*$  is the expected housing yield as horizon goes to infinity.
- **Option value of future extensions.** Before extension, a leasehold has an option to extend in the future. The share of the extension value retained by the leaseholder is summarized by a parameter  $1 - \alpha$ .
- **Empirical simplification.** The paper argues that  $\alpha \approx 1$ : the freeholder captures almost all extension value through the negotiated payment, so the pre-extension option value is close to zero. This makes the post-extension price gain especially informative about long-horizon discounting.

### 3 Models and empirical specifications (all equations + interpretation)

#### 3.1 The long-term housing yield

The price of housing equals the expected discounted present value of its future rental flow. The paper writes the price of one unit of housing as

$$P_t = R_t \mathbb{E}_t \left[ \int_0^\infty \exp \left( - \int_0^s y_t(u) du \right) ds \right],$$

where the forward housing yield is

$$y_t(u) = r_t(u) + \zeta_t(u) - g_t(u).$$

Here  $r_t(u)$  is the forward safe rate,  $\zeta_t(u)$  is the housing risk premium, and  $g_t(u)$  is expected rental growth.

The main long-run object is

$$y_t^* \equiv \lim_{u \rightarrow \infty} \mathbb{E}_t[y_t(u)] = r_t^* + \zeta_t^* - g_t^*.$$

- **Interpretation.**  $y_t^*$  is the expected yield on housing after short-run and medium-run disturbances have dissipated.
- **Equivalent intuition.** It is also the long-run expected rent-price ratio, so it summarizes the valuation the market expects in the long run.

#### 3.2 Pricing a finite-duration leasehold with an extension option

Let  $P_t^T$  denote the price of a leasehold with  $T$  years remaining. The paper writes

$$P_t^T = R_t \int_0^T \exp \left( - \int_0^s y(u) du \right) ds + \max \left\{ 0, (1 - \alpha) R_t \int_T^{T+90} \exp \left( - \int_0^s y(u) du \right) ds + \dots \right\}.$$

- **First term.** Present value of housing services before the lease would expire.
- **Second term.** Option value of extending the lease later, net of the payment to the freeholder.
- **Key institutional restriction.** Because leaseholders must compensate the freeholder for the market value of the extension, the paper sets  $\alpha = 1$  as the benchmark, implying essentially zero pre-extension option value for the leaseholder.

#### 3.3 Difference-in-differences estimand

The paper's first step is to estimate the increase in market value caused by extension. For a treated property  $i$  and a nonextending control  $j$ , the difference-in-differences estimand is

$$\bar{\Delta}_{it} = [\log P_{it}^{T+k} - \log P_{i,t-h}^{T+h}] - [\log P_{jt}^T - \log P_{j,t-h}^{T+h}].$$

The first bracket is price growth for the treated property from purchase to sale. The second is the corresponding price growth for the control property over the same window.

- **Interpretation.**  $\bar{\Delta}_{it}$  is the market-value gain from extension after netting out local price growth common to comparable nonextending properties.
- **What it removes.** General housing-market appreciation, local demand shocks, and any common movements in service flows, provided the parallel-trends assumption holds.

### 3.4 From DiD to the pricing equation

Substituting the leasehold-pricing formula into the difference-in-differences estimand gives

$$\bar{\Delta}_{it} = \log \left( \int_0^{T+k} \exp \left[ - \int_0^s y(u) du \right] ds \right) - \log \left( \int_0^T \exp \left[ - \int_0^s y(u) du \right] ds \right) + \Delta_{t,t-h}(\log R_i - \log R_j).$$

The last term captures differential growth in the service flow between treated and control properties.

- **Parallel trends assumption.** The growth in service flow must be the same for treated and control properties over the experiment window, so the final term vanishes on average.
- **Economic meaning.** Under parallel trends, the post-extension price gain isolates the duration effect rather than changes in housing quality or local demand conditions.

### 3.5 The DiD-NLLS estimator

To estimate  $y_t^*$ , the paper parameterizes the long-run forward housing yield as flat at sufficiently long horizons,  $y(s) = y^*$ . Under this restriction,

$$\bar{\Delta}_{it} = \log (1 - e^{-y^*(T+k)}) - \log (1 - e^{-y^*T}).$$

This equation is estimated by nonlinear least squares. Allowing for time variation gives the period- $t$  estimating equation

$$\bar{\Delta}_{it} = \log (1 - e^{-y_t^*(T_{it}+k)}) - \log (1 - e^{-y_t^*T_{it}}).$$

- **Key identifying variation.** For a given duration  $T$ , the gain from extension is larger when  $y_t^*$  is lower. For a given  $y_t^*$ , the gain is larger when the pre-extension duration is shorter or the extension length  $k$  is larger.
- **Why the flat-yield assumption is reasonable.** The paper does not require the whole yield curve to be flat. What matters is that the forward housing yield has flattened out by the very long durations relevant for the lease-extension comparison.

### 3.6 Why the estimator differences out the short end

A central conceptual point of the paper is that the estimator largely removes the effect of short- and medium-run yields. The reason is that the comparison is between a lease with duration  $T$  and one with duration  $T + k$ , where  $T$  is already very large.

- When  $T$  is large, the difference between the two prices is driven mostly by cash flows between  $T$  and  $T + k$ .

- Therefore large fluctuations in short-end or medium-end yields have only a limited effect on the estimated  $y_t^*$ .
- This is why the method can “look through” monetary tightenings, temporary bubbles, pandemic shocks, and other shorter-run disturbances.

The paper illustrates this numerically by simulating many yield curves with the same long-run limit  $y^*$  but very different short-end shapes, and shows that the implied  $\hat{y}^*(T)$  varies little when  $T$  is large.

### 3.7 Control-group construction using repeat sales

For each treated property, the paper constructs a repeat-sales control index from nonextending properties that:

- do not extend between  $t - h$  and  $t$ ,
- have duration within five years of the treated property,
- are located within the smallest feasible radius  $d$  kilometers, capped at 20 km.

The repeat-sales control is calculated separately for each treated property.

- **Why repeat sales help.** Exact same-time purchases and sales would be too restrictive and would sharply reduce the usable sample.
- **Why this does not change identification.** The identifying assumption remains the same: treated and control properties must have similar service-flow growth over the experiment window.

### 3.8 Validation regression using rent growth

Because rents proxy for housing service flow, the paper directly tests parallel trends in rents with

$$\Delta \log R_{i,j,t',t''} = \alpha_{i,t',t''} + \beta_1 \text{Extension}_j + \varepsilon_{i,j,t',t''},$$

where  $\alpha_{i,t',t''}$  are experiment-by-rent-years fixed effects.

- **Interpretation.** If  $\beta_1 = 0$ , treated and control properties exhibit the same rent growth, supporting the service-flow parallel-trends assumption.
- **Why this is powerful.** Unlike sale prices, rents are directly tied to service flow, so this is a close test of the underlying identifying assumption.

### 3.9 Cross-sectional regressions for climate risk and supply elasticity

After estimating local values of  $y^*$ , the paper studies two cross-sectional patterns.

First, for climate risk,

$$y_\ell^* = a + b_1 \text{FloodRisk}_\ell + b_2 \text{SubsidenceRisk}_\ell + u_\ell,$$

where  $\ell$  indexes local authorities.

Second, for housing supply elasticity, the paper studies long-run changes in local yields,

$$\Delta y_\ell^* = c + d_1 \text{RefusalRate}_\ell + d_2 \text{ShareDeveloped}_\ell + e_\ell,$$

and also instruments the refusal rate using the post-2002 change in planning delay rates developed by Hilber and Vermeulen.

- **Climate-risk interpretation.** Higher  $y^*$  in risky areas is consistent with higher long-run housing risk premia.
- **Supply-elasticity interpretation.** If housing supply is more inelastic, a fall in long-run discount rates translates more strongly into higher valuations and lower  $y^*$ .

## 4 Identification and instruments (what makes the estimates credible)

### 4.1 What fails in naive approaches

- **Current prices mix many horizons.** A spot rent-price ratio reflects short-run, medium-run, and long-run factors simultaneously, so it cannot isolate the expected long-run yield.
- **Cross-sectional duration comparisons are fragile.** Different lease durations may be attached to different kinds of properties, neighborhoods, and service flows, so pure cross-sectional comparisons risk omitted-variable bias.
- **Structural long-run yield estimates are model-dependent.** Existing methods for  $r^*$  or long-run yields often rely on large macro structures that can be misspecified, especially during unusual episodes like the pandemic.

### 4.2 What identifies the model in the new approach

- **Within-property duration variation.** Lease extension changes the maturity structure of the same underlying property.
- **Difference-in-differences design.** Comparing treated price growth with nearby nonextending controls removes common local housing-market movements.
- **Long horizon.** Because the relevant cash flows are decades in the future, the estimator is naturally focused on the long end of the housing yield curve.
- **Legal structure of extension pricing.** Since the leaseholder must compensate the freeholder for the extension value, the pre-extension embedded option is close to valueless for the leaseholder, simplifying the pricing comparison.

### 4.3 Why the parallel-trends assumption is plausible

- **Motivation from behavior.** Lease extension is mainly a valuation-management decision: owners extend so the property can be sold at a higher price or avoid becoming too short-dated.
- **No pre-trends in prices.** The event-study analysis shows no differential price growth between treatment and controls before extension.

- **No differential renovation.** Text analysis of listings and changes in bedrooms, bathrooms, living rooms, and floor area show no evidence that treated properties are improved more than controls around extension.
- **No differential rent growth.** Rents evolve similarly for treated and control properties both inside and outside the experiment window.
- **Observed hedonic balance.** The treated and control groups are similar on the observed characteristics available in Rightmove and Zoopla data.

#### 4.4 Why the repeat-sales control strengthens the design

- It avoids the “forbidden comparisons” problem by using only nonextending controls.
- It expands the feasible control set greatly relative to exact same-time sales matching.
- It lets the analysis remain local in duration and geography while using many more natural experiments.

#### 4.5 What the instrument does in the supply-elasticity exercise

- **Problem.** The observed major-construction refusal rate is endogenous because developers may selectively avoid applying in places where approval is unlikely.
- **Instrument.** The paper uses the change in application delay rates after a 2002 reform. More restrictive local authorities had more reason to reduce delays after the reform.
- **Role.** This IV is not needed for the core estimate of  $y_t^*$ ; it is used only to strengthen the interpretation of the cross-sectional relationship between housing-supply inelasticity and the long-run fall in  $y^*$ .

#### 4.6 Relation to the literature

- **Relative to Giglio, Maggiori, and Stroebel (2015).** That paper used cross-sectional variation in lease durations to estimate long-run housing discounting. This paper extends the idea by exploiting within-property lease extensions and real-time updating.
- **Relative to structural  $r^*$  estimation.** The paper offers a market-based, microdata-based alternative that avoids placing a full macro structure on the data.
- **Relative to housing macro.** The results connect housing valuations, supply elasticity, climate risk, and secular movements in long-run discounting.

## 5 Tables and figures

### 5.1 Figure 1: Time series of the expected long-term housing yield

Read:

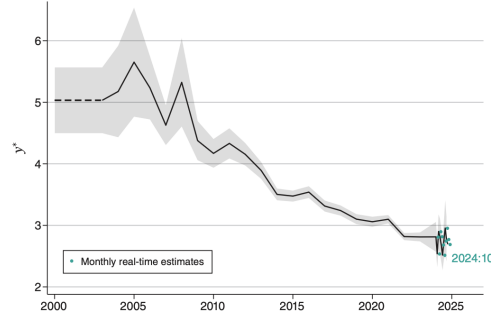


FIGURE 1. TIME SERIES OF THE EXPECTED LONG-TERM HOUSING YIELD

Notes: The figure shows estimates of  $y^*$  over time for the full sample of lease extensions. Estimates for 2000–2003 are pooled and are reported in the dashed black line. Yearly estimates of  $y^*$  for 2004–2023 are plotted in the solid black line. Monthly estimates for January 2024 through October 2024 are also reported. The shaded area shows 95 percent confidence intervals for the estimates. Standard errors are heteroskedasticity robust.

- The headline result is a large secular decline in  $y^*$ : from about **5.2%** in the early 2000s to about **2.7% in 2024**.
- The decline begins around the Great Recession rather than during the pandemic.
- The pandemic generates little movement in the long-run housing yield, even though shorter-run housing and financial-market conditions were highly volatile.
- The confidence bands tighten over time as the number of lease-extension natural experiments increases.

## 5.2 Table 1 and Figure 2: Extension distribution and the timing notation

| Extension amount | 90     | 700+   | Other  | Total   |
|------------------|--------|--------|--------|---------|
| 2000–2005        | 819    | 2,793  | 2,806  | 6,418   |
| 2006–2010        | 3,416  | 7,107  | 5,364  | 15,887  |
| 2011–2015        | 11,753 | 15,806 | 10,755 | 38,314  |
| 2016–2020        | 16,996 | 19,434 | 12,199 | 48,629  |
| 2021–2023        | 13,281 | 14,263 | 6,822  | 34,366  |
| Total            | 46,265 | 59,403 | 37,946 | 143,614 |

Notes: The table reports the number of extended leases that have transaction data for each time period. The first column includes 90 year extensions, the next column includes 700+ year extensions, and the third column includes others, which are almost all non-90 under 200 year extensions. The first column denotes the time of sale after extension.

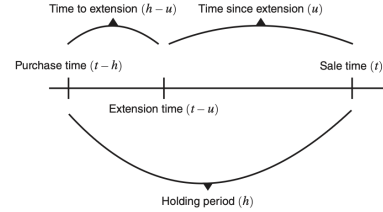


FIGURE 2. DIAGRAM OF EXTENSION TIME

Notes: The figure is a diagrammatic representation of the notation we will use in the paper. We say a property is purchased at time  $t - h$ , sold at time  $t$  and held for an amount  $h$  of years. We say that a property extends at time  $t - u$ , where  $0 < u < h$ .

### Read:

- The treatment sample contains **143,614** lease extensions with transaction prices before and after extension.
- The most common extension lengths are **90 years** and **700+ years**, which gives identifying variation at different long maturities.
- Figure 2 clarifies the experiment structure: purchase at  $t - h$ , extension at  $t - u$ , sale at  $t$ , and remaining duration  $T$  absent extension.
- This timing diagram is crucial because the identification comes from how extension changes the maturity of the same underlying housing asset.

### 5.3 Figures 3 and 4: Why the estimator isolates the long end

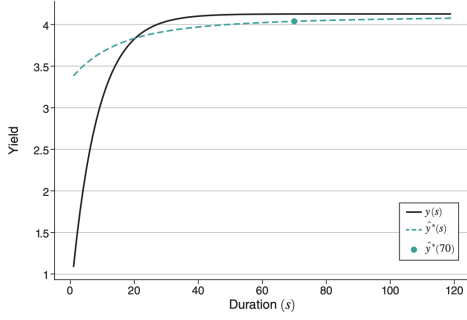


FIGURE 3. A PARAMETERIZATION OF THE HOUSING YIELD CURVE

Notes: The black line presents one parameterization of the forward yield curve,  $y(s)$ , which is chosen so that its shape matches the forward curve implied by the 1 year gilts, the 10 year gilts, and the 25 year gilts, averaged over the 2010–2020 period.  $\hat{y}^*(s)$  is our estimator of  $y^*$  for each  $T$ , described below.

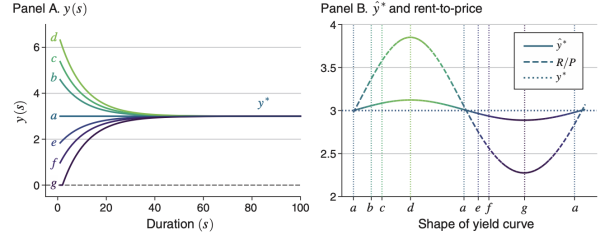


FIGURE 4. “DIFFERENCING OUT” THE SHORT END

Notes: The figure illustrates the effect of fluctuations at the low end of the yield curve on  $\hat{y}^*$  and the rent-to-price ratio, the rent-to-price ratio. Panel A presents several instances of the yield curve,  $y(s)$  over time. Panel B indicates the estimates  $\hat{y}^*$  and the rent-to-price ratio at each of these instances. For instance, point  $d$  on the right panel corresponds to the true value of  $y^*$ , the estimated value of  $y^*$ , and the rent-to-price ratio; given a yield curve  $d$  on the left panel. The rent-to-price ratio is estimated such that  $P_d^R/R_d = \int_0^\infty e^{-\int_0^s y(s) ds} ds \equiv \frac{1}{R/P}$ .

Read:

- Figure 3 shows numerically that the estimator  $\hat{y}^*(T)$  tracks the true long-run yield closely once the pre-extension duration  $T$  is large.
- Figure 4 shows that even large movements in the short end of the yield curve have only small effects on the estimated  $y^*$ .
- By contrast, a simple rent-price ratio would move much more with the short end.
- These figures are the paper’s main intuition for why lease extensions recover a long-run object rather than a short-run one.

### 5.4 Table 2, Table 3, and Figure 5: Validation of parallel trends

|                          | Renovation rate<br>(1) | $\Delta$ Bedrooms<br>(2) | $\Delta$ Bathrooms<br>(3) | $\Delta$ living rooms<br>(4) | $\Delta$ floor area<br>(5) |
|--------------------------|------------------------|--------------------------|---------------------------|------------------------------|----------------------------|
| Extension                | −0.001<br>(0.001)      | −0.002<br>(0.001)        | 0.000<br>(0.000)          | −0.001<br>(0.000)            | −0.002<br>(0.001)          |
| Experiment fixed effects | ✓                      | ✓                        | ✓                         | ✓                            | ✓                          |
| Control mean             | 0.091                  | 0.04                     | 0.002                     | 0.001                        | 0.022                      |
| Observations             | 153,400                | 43,612                   | 33,078                    | 31,934                       | 34,564                     |
| N. experiment            | 76,700                 | 21,806                   | 16,539                    | 15,967                       | 17,282                     |

Notes: The table reports renovation rates for extended properties relative to their nonextended counterparts for all experiments. For each experiment we have two observations indexed with  $j$ : the renovation rate for the extended property, and the mean renovation rate across all control properties. We run the following regression,  $X_{ij} = \alpha_i + \beta_1(\text{Extended})_i + \epsilon_{ij}$  for renovation measures  $X_j$  where  $\alpha_i$  are experiment fixed effects. The renovation rate for treated properties in column 1 is based on the post-extension sale listing. The renovation rate for columns 2–5 is based on change in hedonic characteristics before and after extension time. Standard errors are heteroskedasticity robust. Standard errors in parentheses.

|                                              | $\Delta \log(\text{Rent})$ |                     |                     |
|----------------------------------------------|----------------------------|---------------------|---------------------|
|                                              | (1)                        | (2)                 | (3)                 |
| Extension                                    | 0.0007<br>(0.0006)         | −0.0006<br>(0.0006) | −0.0003<br>(0.0007) |
| Experiment $\times$ rent years fixed effects | ✓                          |                     | ✓                   |
| Experiment fixed effects                     |                            | ✓                   |                     |
| Annualized RSI                               |                            |                     | ✓                   |
| Observations                                 | 3,943,474                  | 36,410              | 74,266              |
| N. Experiment                                | 16,569                     | 18,205              | 18,887              |

Notes: The table reports rent growth during the experiment window of  $t - h$  to  $t$  for extended properties relative to control properties. The first column is the estimated coefficient from equation (7). We use mean annualized rent growth as the LHS variable, and do not control for time fixed effects. The third column uses a rent repeat sales index. Standard errors are heteroskedasticity robust. Standard errors in parentheses.

Read:

- Table 2 shows essentially **no difference in renovation rates** between extended properties and their controls.
- Table 3 shows **no meaningful difference in rent growth** between treated and control properties within the experiment window.

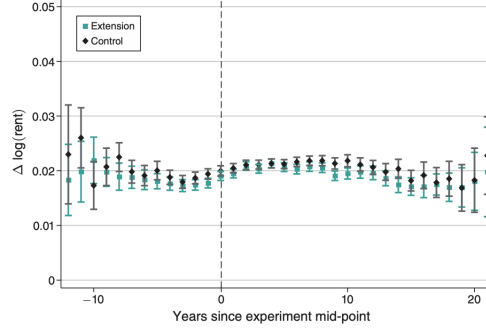


FIGURE 5. RENT GROWTH AT LONG HORIZONS

Notes: The figure presents annual rent growth at all horizons for treated properties relative to a repeat sales index of rental prices for controls. We interpolate across years when rental listings are not consecutive and plot rent growth against time since the experiment window. The x-axis variable is the mid-point between the purchase and sale of the treated property,  $(t + (t + h))/2$ . Error bars represent 95 percent confidence intervals.

- Figure 5 extends this logic to longer horizons and again finds similar rent-growth paths for treated and control properties.
- Together these are the paper's strongest direct validation exercises for the identifying assumption that service-flow growth is parallel across treatment and controls.

## 5.5 Figure 6 and Figure 7: Event-study and NLLS intuition

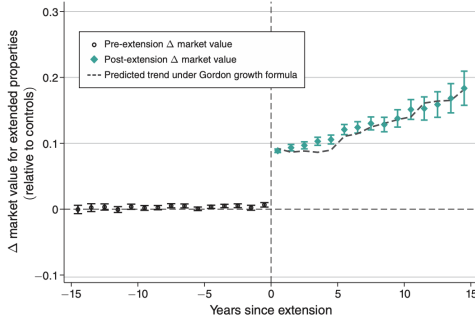


FIGURE 6. EVENT STUDY REPRESENTATION OF LEASE EXTENSION

Notes: The figure presents an event study representation of a lease extension. On the positive x-axis we plot in blue diamonds the mean difference in price change between time  $t - h$  and  $t - h + \Delta$ , against time since extension,  $u$ . On the negative x-axis we plot in black circles the mean difference in price change between time  $t - h - h'$  and  $t - h$  for extended versus control properties.  $t - h - h'$  is the time of the nearest transaction before  $t - h$ . The dotted line plots the predicted price gain based on mean duration and  $y^*$  at each time. The sample includes all extended properties. 95 percent confidence intervals are shown.

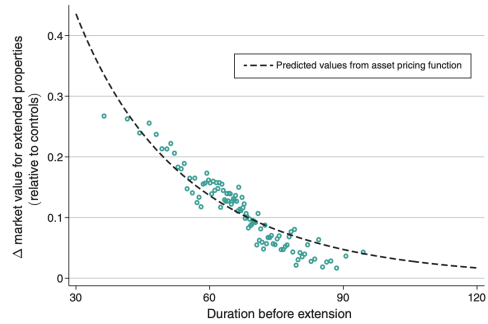


FIGURE 7. DURATION BEFORE EXTENSION VERSUS PRICE GAIN AFTER EXTENSION

Notes: The figure is a binscatter of our difference-in-difference estimator against duration before extension,  $T$ , with 100 bins. The sample includes leases that were extended for 90 years. The black line shows fitted estimates of equation (6).

### Read:

- Figure 6 shows a clean post-extension jump in market value with **no pre-trend**. The gain is about **0.1 log points** one year after extension and rises toward **0.2 log points** after ten years.
- The dashed predicted line from the asset-pricing formula tracks the event-study pattern closely.
- Figure 7 shows the gain from a 90-year extension is much larger when the property had a shorter remaining duration before extension.
- Quantitatively, the gain is around **7%** when the lease has about 90 years remaining and above **30%** when it has around 40 years remaining.

## 5.6 Table 4 and Figure 8: Average y-star and the flat long-run yield curve

TABLE 4—ESTIMATED  $y^*$

|                            | Constant $y^*$  | Flexible $y^*$  |                 |                 |                 | Constant $y^*$      |
|----------------------------|-----------------|-----------------|-----------------|-----------------|-----------------|---------------------|
|                            | $k = 90$<br>(1) | $T = 50$<br>(2) | $T = 60$<br>(3) | $T = 70$<br>(4) | $T = 80$<br>(5) | $k \geq 700$<br>(6) |
| $y^*$                      | 3.51<br>(0.023) | 3.55            | 3.51            | 3.48            | 3.44            | 3.50<br>(0.019)     |
| Observations               | 42,429          |                 |                 |                 |                 | 54,581              |
| $t$ -stat (700+ versus 90) |                 |                 |                 |                 |                 | −0.32               |

Notes: Column 1 presents estimates of  $y^*$  from equation (6) for all extensions. Columns 2–6 present estimates of  $y^*(T+k)$  for the range of durations in our sample, where we parameterize  $y^*$  linearly as a function of  $T+k$ ,  $y^*(T+k) = \alpha + \beta \cdot (T+k)$ . We estimate  $y^*$  using a version of  $\Delta_{\Delta y}$  that has been residualized on year  $\times$  local authority fixed effects. Standard errors are heteroskedasticity robust. Standard errors in parentheses

Read:

- Table 4 estimates an average full-sample long-run housing yield of about **3.51%**.
- Allowing the estimate to vary flexibly with pre-extension duration barely changes the result: values remain in a narrow range around **3.44%–3.55%**.
- Estimating the model on **700+ year extensions** gives essentially the same  $y^*$ , which supports the idea that the forward housing yield is quite flat at very long horizons.
- Figure 8 shows the market-value gain from 90-year extensions rises by about **10 percentage points** between the start and end of the sample, visually foreshadowing the large decline in  $y^*$  over time.

## 5.7 Figure 9: Monthly real-time estimates of y-star

Read:

- The paper can update  $y^*$  in real time because there are now close to **1,000 lease extensions per month**.
- The monthly series confirms the same broad pattern as the annual series: a major decline after the Great Recession and little movement during the pandemic.
- The figure also shows that after 2022 real government-bond forward rates rise sharply while  $y^*$  remains stable.
- The implied message is that recent increases in medium-term rates need not reflect a rise in the very long-run housing yield or in  $r^*$ .

## 5.8 Figure 10: Stability relative to the cross-sectional approach

Read:

- The cross-sectional lease-duration estimator is highly sensitive to how hedonic controls are specified.
- Depending on controls, the cross-sectional estimate ranges from about **1.78% to above 10%**.

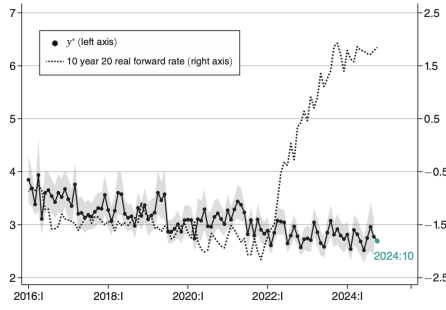


FIGURE 9. MONTHLY ESTIMATES OF  $y^*$  AND RATE OF RETURN ON GOVERNMENT BONDS

Notes: The solid line presents  $y^*$  estimated at a monthly frequency using all extensions. The line is plotted from January 2016 to October 2024. 95 percent confidence interval is shaded.

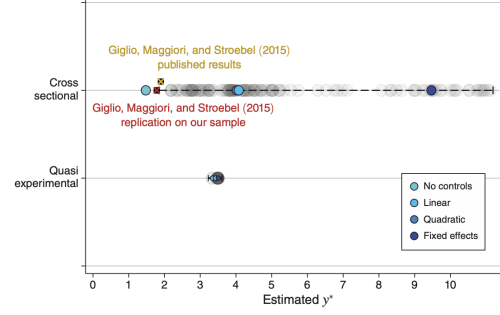


FIGURE 10. STABILITY OF  $y^*$ : CONTROLLING FOR OBSERVABLES

Notes: The figure presents estimates of  $y^*$  under various choices of hedonic controls. For the cross-sectional estimates, we estimate  $\log P_t^f - \log P_t^r = \log(1 - e^{-T})$  by NLLS, where  $P_t^r$  is the price of a freehold transacted in the same quarter and local authority as a  $T$  duration leasehold. For the quasi-experimental estimates, we follow the methodology described in Section III. For each methodology, we perform over 100 estimations, controlling for different combinations of hedonic characteristics. We indicate in various shades of blue four important sets of controls: no controls, linear controls, quadratic controls, and the full set of hedonic fixed effects. The gold cross presents the  $y^*$  estimate from Giglio, Maggiori, and Stroebe (2015), and the red cross presents our replication of their estimate, using the full data from 2000–2023.

- The quasi-experimental estimate from lease extensions stays tightly clustered around the baseline value.
- This is one of the clearest demonstrations in the paper that within-property duration variation is much more robust than pure cross-sectional duration comparisons.

## 5.9 Figure 11: Comparing other asset yields to $y^*$

Read:

- The long-run trends in UK and global real forward safe rates and in rent-price ratios line up closely with the decline in  $y^*$ , especially before 2020.
- This suggests that the fall in  $y^*$  is not purely a housing-specific phenomenon.
- The common movement points toward a broader decline in long-run discount rates, plausibly  $r^*$ .
- The post-2020 gap between medium-term safe rates and  $y^*$  is then especially informative about whether the recent rate increase is temporary or permanent.

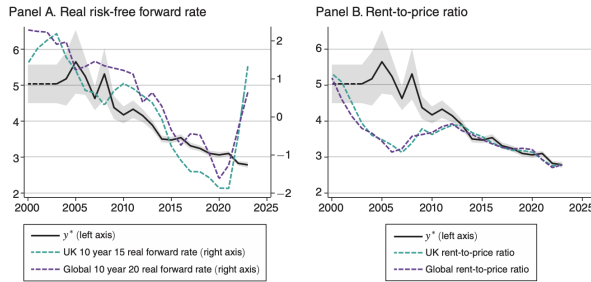


FIGURE 11. COMPARING OTHER ASSET YIELDS TO  $y^*$

Notes: The figures present yields for other assets over the 2003–2023 period. Panel A plots the 10-year-15 real forward rate for the United Kingdom, using data on real forward curves from the Bank of England (Bank of England 1970). It also plots the weighted average 10-year-20 forward rate for nine OECD countries for which data is available from Global Financial Data (2024). The average is weighted by each country's GDP. Since data on inflation-linked bonds is not available for many countries, we define the real forward rate as the nominal forward rate minus 3 percent expected inflation. Panel B plots the UK rent-to-price ratio and the weighted average rent-to-price ratio for 22 OECD countries for which it is available (OECD 2023). The rent-to-price ratio is calculated from a rent-to-price index published by OECD, and normalized to the 2023  $y^*$  level.

TABLE 5—DETERMINANTS OF LONG-RUN RISK

|                 | $y^*$          |                |
|-----------------|----------------|----------------|
|                 | (1)            | (2)            |
| Flood risk      | 0.19<br>(0.07) |                |
| Subsidence risk |                | 0.33<br>(0.11) |
| Observations    | 224            | 224            |
| $R^2$           | 0.05           | 0.10           |

Note: Standard errors in parentheses.

## 5.10 Table 5, Table 6, and Figure 12: Climate risk and supply elasticity

|                      | $\Delta y^*$    |                 |                 |                | Refusal rate     | $\Delta y^*$      |                  |
|----------------------|-----------------|-----------------|-----------------|----------------|------------------|-------------------|------------------|
|                      | OLS<br>(1)      | OLS<br>(2)      | OLS<br>(3)      | OLS<br>(4)     | 1st stage<br>(5) | IV<br>(6)         | IV<br>(7)        |
| Refusal rate         | −4.29<br>(1.25) | −6.35<br>(1.27) | −6.87<br>(1.36) |                |                  | −17.94<br>(11.19) | −13.74<br>(5.44) |
| Share developed      |                 | −1.63<br>(0.32) | −1.82<br>(0.54) |                |                  |                   | −2.38<br>(0.72)  |
| Change in delay rate |                 |                 |                 | 2.52<br>(0.64) | −0.17<br>(0.04)  |                   |                  |
| Region fixed effects |                 |                 | ✓               |                |                  |                   |                  |
| Observations         | 224             | 224             | 224             | 224            | 224              | 224               | 224              |
| $R^2$                | 0.11            | 0.30            | 0.34            | 0.11           | 0.08             | −1.05             | 0.00             |

*Notes:* The table presents cross-sectional regressions of the long-run change in  $y^*$  against local authority level measures of supply-side constraints in the United Kingdom. The long-run change in  $y^*$  is defined as the difference between post-2022  $y^*$  and pre-2009  $y^*$  (inclusive). Columns 1 through 4 present OLS regressions of  $\Delta y^*$  against three measures of supply elasticity: the average refusal rate of major construction applications, the share of developable land developed, and the change in delay rate following the 2002 policy reform, all of which are discussed in the main text. In column 3 we also include region fixed effects. Column 6 and 7 present IV regressions, where the refusal rate is instrumented using the change in delay rate after 2002. Standard errors in parentheses.

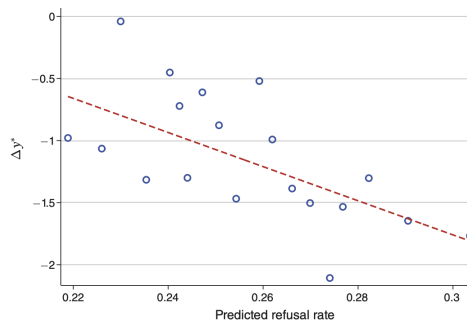


FIGURE 12. THE DECLINE IN  $y^*$  DEPENDS ON HOUSING SUPPLY ELASTICITY

*Note:* The figure presents a binscatter with 20 bins of the long-run change in  $y^*$ ,  $\Delta y^*$ , against the predicted values from the first stage of the IV in column 6 of Table 6.

### Read:

- Table 5 shows that local authorities with greater exposure to **flood risk** or **subsidence risk** have higher values of  $y^*$ , consistent with higher long-run housing risk premia.
- Table 6 shows that local authorities with more restrictive housing supply—measured by planning refusal rates and land-development constraints—experience larger long-run declines in  $y^*$ .
- The IV estimates using the post-2002 planning-delay reform point in the same direction and reinforce the supply-elasticity interpretation.
- Figure 12 visualizes the result: more supply-constrained places show larger negative long-run changes in  $y^*$  when long-run demand pressure rises.

## 6 Short summary

- **Conclusion.** The paper identifies the expected long-run housing yield from the price gain generated by leasehold extensions, using a difference-in-differences design plus discounted-cash-flow logic.
- **Main identification insight.** By comparing the same property's value before and after a large maturity extension, relative to nearby nonextending leaseholds, the method strips out service-flow heterogeneity and most short-end yield fluctuations.
- **Main empirical lesson.** Long-run expected housing yields in the United Kingdom fell dramatically after the Great Recession, remained low through the pandemic and into 2024, are higher in climate-risky regions, and declined more in supply-inelastic places.

## 7 Conclusion

### 7.1 Core conclusion (what the paper establishes)

- The paper shows that leasehold extensions provide a clean natural experiment for identifying very long-run housing yields.

- It documents a large secular fall in the UK expected long-run housing yield, from roughly 5.2% in the early 2000s to about 2.7% by 2024.
- It also shows that this measure can be updated in real time with public administrative data, making it useful for ongoing macro-financial monitoring.

## 7.2 Economic intuition

- A lease extension gives the owner an additional 90 years or more of housing services. If long-run discount rates are low, those distant cash flows are very valuable, so extension raises the market price a lot.
- Because the same property is compared before and after extension, and relative to nearby controls, the estimator is much less sensitive to unobserved housing quality than a cross-sectional duration comparison.
- Since the extra cash flows arrive far in the future, the design naturally focuses on the long end of the housing yield curve rather than the short end.

## 7.3 Takeaway

This paper is best read as a measurement paper with unusually sharp identification and important macro implications. Its central message is that long-run expected yields can be learned from housing microdata when the maturity structure of the same asset changes in a quasi-experimental way. The resulting series for  $y_t^*$  is useful both as a direct object—the market’s expected long-run housing yield—and as a diagnostic for broader questions about secular discount-rate declines, the persistence of house-price booms, the role of supply elasticity, and the pricing of long-run climate risk.